DEPARTMENT OF ELECTRONICS AND ELECTRICAL ENGINEERING

Indian Institute of Technology Guwahati

Average Current Mode Control of Flyback DC–DC Converter

Project Report

Master of Technology (M.Tech)

Power Engineering

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Abstract

This report presents the systematic analysis, small-signal modeling, and closed-loop compensator design for a non-inverting PWM flyback DC–DC converter operating in Continuous Conduction Mode (CCM) under Average Current Mode Control (ACMC). The converter is designed to regulate a 5 V output from a 12 V supply at a switching frequency of 100 kHz, with a turns ratio of 3:1 and a magnetizing inductance of 200 μH. The control-to-output and current-loop plant transfer functions are obtained and their frequency-domain characteristics, including the Right-Half-Plane (RHP) zero, are analyzed. Inner current loop and outer voltage loop compensators are designed using classical frequency-domain methods, ensuring adequate gain and phase margins. The compensated loop gains are verified through Bode plot analysis. Simulation results validate the theoretical predictions for reference tracking, line regulation, and load transient response. The overall design demonstrates effective closed-loop regulation and satisfactory dynamic performance, while the ACMC strategy provides accurate current control, improved transient response, and stable converter operation.

1. Introduction

1.1 Background and Motivation

The flyback DC–DC converter is one of the most extensively deployed isolated power converter topologies in modern power electronics. Its structural simplicity—achieved by combining the functions of an energy storage inductor and an isolation transformer in a single coupled magnetic component—makes it particularly attractive for low-to-medium power applications ranging from consumer electronics to industrial power supplies and telecommunication systems. The isolated nature of the topology allows it to provide galvanic separation between the input and output, which is a regulatory and safety requirement in numerous applications.

The performance of a flyback converter is critically dependent on the control strategy employed. Voltage-mode control (VMC), while straightforward to implement, suffers from poor dynamic response to input voltage perturbations and requires an outer voltage loop with a sufficiently high crossover frequency to achieve acceptable transient performance. Average Current Mode Control (ACMC) overcomes these limitations by introducing a dedicated inner current loop that regulates the average inductor current directly. The inner loop, operating at a higher bandwidth than the outer voltage loop, provides inherent input voltage feedforward, improved noise immunity, and simplified outer loop design.

1.2 Problem Statement

The flyback converter is widely used, but designing a stable closed-loop controller in continuous conduction mode (CCM) is challenging. In CCM, the converter has a right-half-plane (RHP) zero, which limits how fast the voltage control loop can respond. To maintain stability, the voltage loop bandwidth must be kept much lower than the RHP zero frequency, resulting in slower transient response.

In average current mode control (ACMC), the system uses two control loops: an inner current loop and an outer voltage loop. These two loops must be designed with sufficient bandwidth separation to avoid interaction and ensure stable operation.The main objective of this project is to derive the required transfer functions and design the two-loop compensator to achieve stable operation, proper regulation, and acceptable dynamic performance.

2. CCM Flyback Converter and System Description

2.1 Converter Specifications

The flyback converter analyzed throughout this report is designed for the following set of specifications, Table 1 summarizes the complete set of design parameters and component values.

Table 1: Flyback Converter Design Specifications and Component Values

Parameter	Symbol	Value
Input Voltage	V_I	12 V
Output Voltage	V_O	5 V
Load Resistance	R_L	20 Ω
Switching Frequency	f_s	100 kHz
Magnetizing Inductance	L_m	200 µH
Turns Ratio (N1/N2)	n	3
Duty Cycle (designed)	D	0.58
IGBT On-Resistance	r_DS	0.5 Ω
Diode On-Resistance	R_F	0.025 Ω
Capacitor ESR	r_C	10 mΩ
Primary Winding Resistance	r_T1	0.2 Ω
Secondary Winding Resistance	r_T2	0.1 Ω
averaged winding resistance	r	0.8785

Based on these specifications, the operating point quantities are summarized in Table 2. The duty cycle **D = 0.58** was computed from the DC voltage transfer function $\mathbf{M\_V_{dc} = D / [n(1 - D)]}$, The equivalent averaged winding resistance **r = 0.8785 Ω** was obtained by applying the principle of conservation of energy to lump all parasitic resistances into the inductor branch.

Quantity	Value
Input Current (I_in)	0.14 A
Output Current (I_O)	0.25 A
Inductor Current (I_L)	0.23 A
Switch Current (Isw)	0.14 A
Voltage Transfer Function (M_V_{dc})	0.45

Table 2: Computed DC Operating Point

2.2 Flyback Converter Circuit Diagram

The non-inverting PWM flyback DC–DC converter circuit consists of a power IGBT S , an output diode Do, a high-frequency transformer with turns ratio **n = N1/N2**, a filter capacitor **C**, and a load resistor $\mathbf{R_L}$. The transformer is modeled by an ideal transformer in series with the magnetizing inductance **Lm**, which stores the magnetic energy during the switch ON interval and transfers it to the load during the switch OFF interval. The leakage inductances are assumed to be zero, and the coupling coefficient is taken as unity.

During the ON interval ($0 < t \leq DT$), the IGBT conducts and energy is stored in **Lm** from the input source $\mathbf{V_I}$. The diode **Do** is reverse-biased. During the OFF interval ($DT < t \leq T$), the IGBT turns OFF, the inductor current commutates through the diode **Do**, and energy is delivered to the load. The equivalent circuit including the magnetizing inductance **Lm** is shown in Figure 2.1.

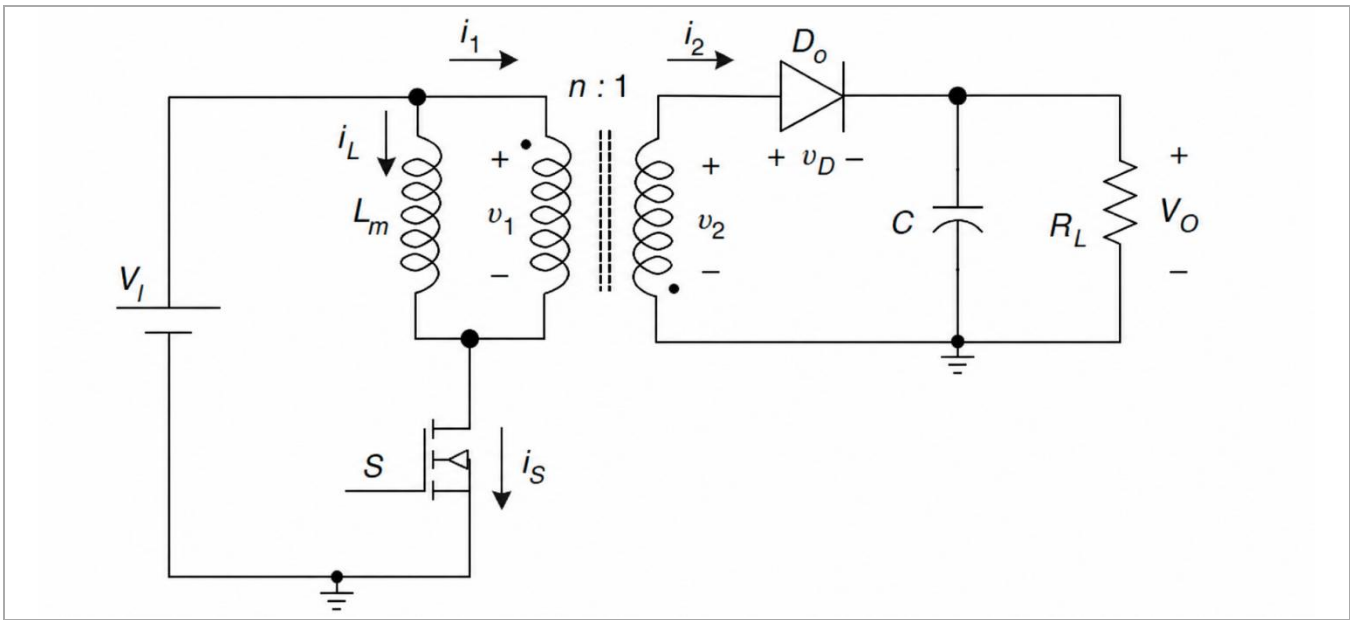


Figure 2.1: *Equivalent Circuit of the Flyback Converter with Magnetizing Inductance Lm.*

2.3 Inductor Current Waveforms (CCM)

In Continuous Conduction Mode, the magnetizing inductance current iL never reaches zero during the switching period. During the ON interval, the current rises linearly with a slope of VI / Lm. During the OFF interval, the current falls linearly with a slope of − nVo / Lm. The peak-to-peak ripple current and the average inductor current are determined by the duty cycle, input voltage, output voltage, and inductance value.

For the design under consideration, the DC inductor current is **IL = 0.231 A**. The waveform comprises a triangular ripple of **0.34** superimposed on this DC value.

The switch current i_S tracking the inductor current during the ON interval and being zero during the OFF interval. The diode current i_D flows during the OFF interval with a peak value determined by the reflected inductor current.

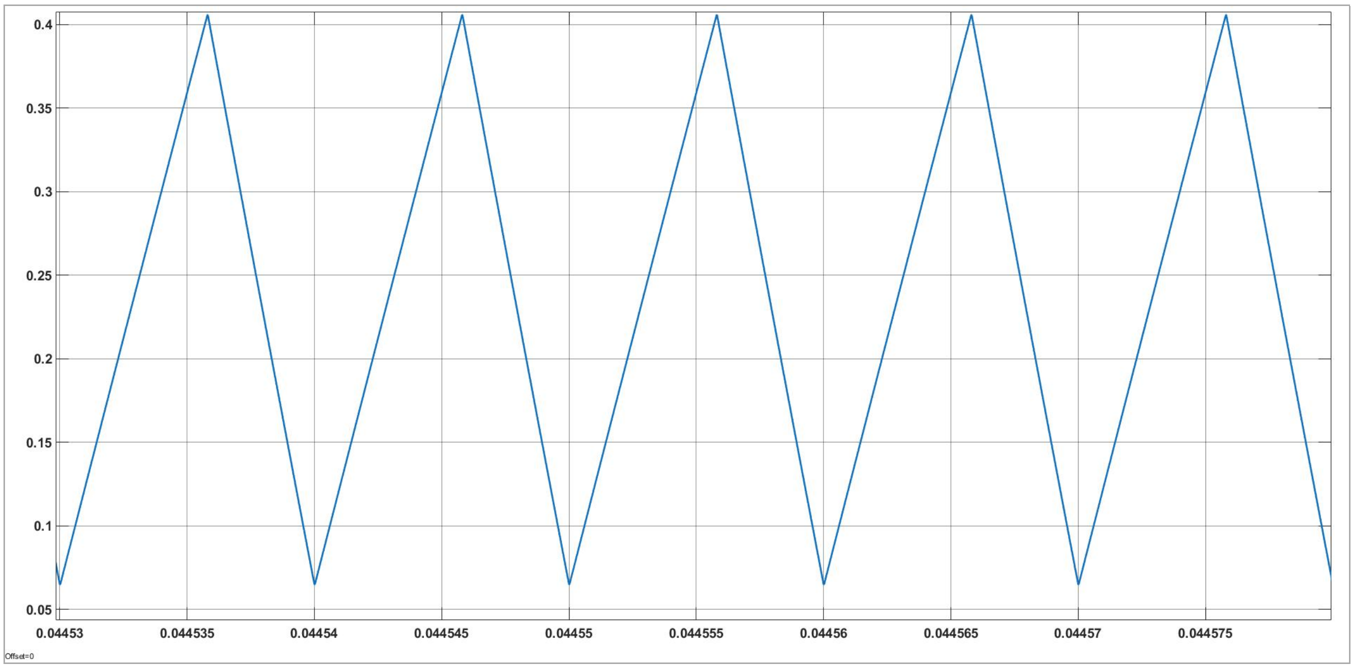


Figure 2.3: *Magnetizing Inductor Current Waveform in CCM Operation.*

2.4 Capacitor Voltage / Output Ripple Characteristics

The output filter capacitor C = 150 μF charges and discharges with the ripple component of the diode current during each switching period. The output voltage ripple is primarily determined by the capacitor ESR rC = 10 mΩ and the capacitance value C. The peak-to-peak output voltage ripple Δv_O can be expressed in terms of the ripple current through the capacitor and its ESR. For the given design, the output ripple (ie 0.02) is small due to the low ESR value.

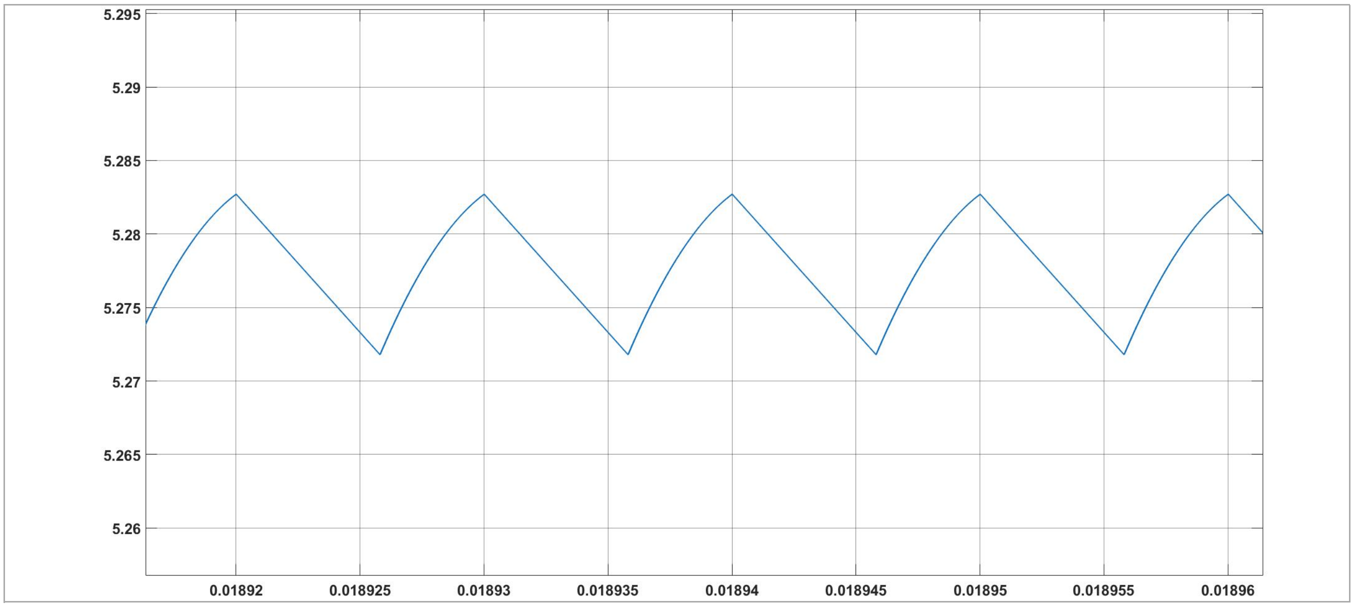


Figure 2.5: *Output Capacitor Current and Voltage Ripple in Steady-State CCM Operation.*

2.5 Switch Current and Voltage Waveforms

The IGBT switch S carries the inductor current i_L during the ON interval and is subjected to a blocking voltage of VI + nVo during the OFF interval. The DC switch current Isw = 0.14 A. The switch voltage stress and current stress determine the IGBT device ratings.

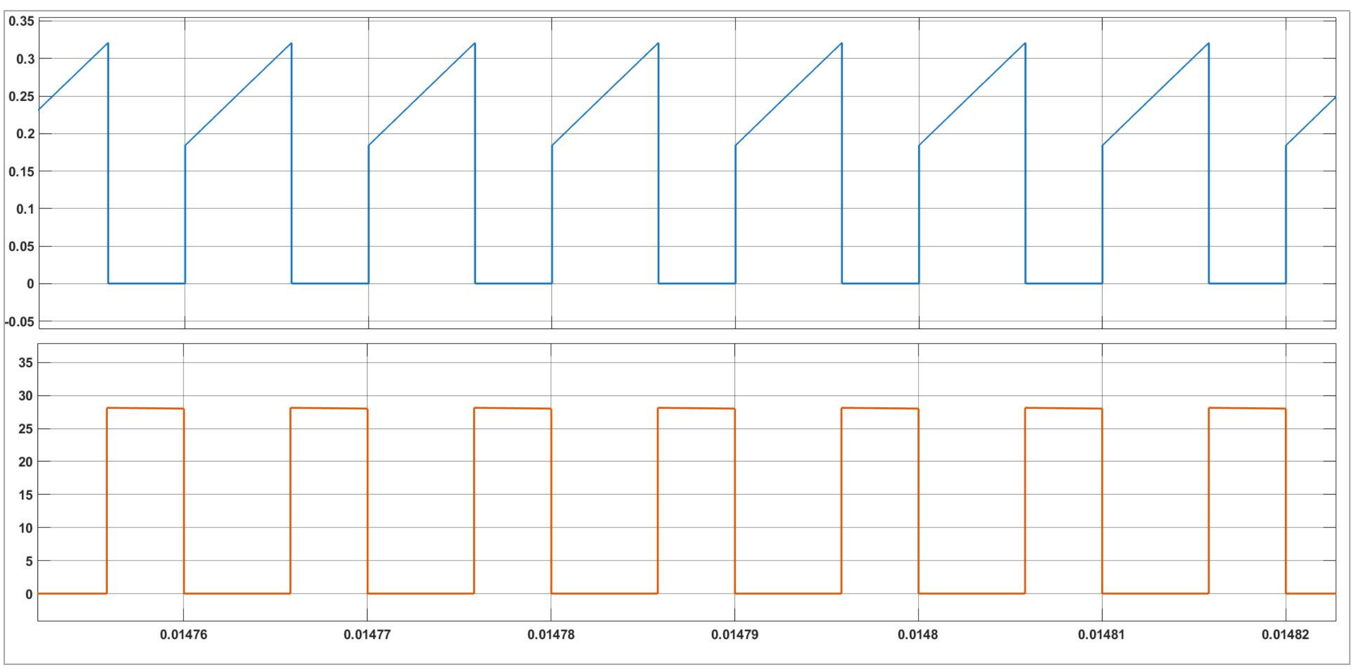


Figure 2.6: *IGBT Switch Current and Voltage Waveforms in CCM.*

2.6 Diode Voltage and Current Waveforms

The output diode Do conducts during the OFF interval, carrying the reflected inductor current from the secondary side of the transformer. The diode is reverse-biased during the ON interval at a voltage of V_I/n + V_O = 9 V. The DC diode voltage is −VD = −5.4 V. The diode current peak equals the peak inductor current reflected by the turns ratio.

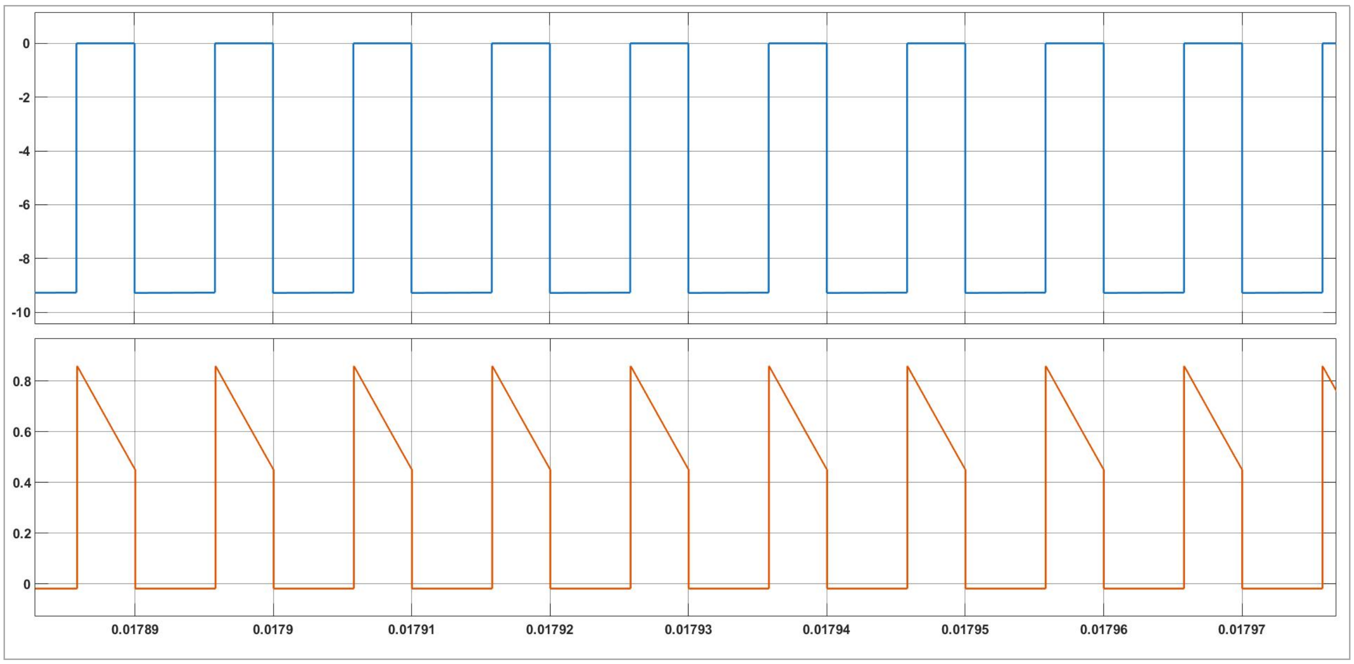


Figure 2.7: *Output Diode voltage and current Waveforms in CCM.*

3. Small-Signal Modeling of the Flyback Converter

Small-signal modeling is the foundation of classical frequency-domain control design. The DC operating point of the converter is perturbed by small-amplitude sinusoidal excitations, linearizing the nonlinear switching equations. The resulting linear time-invariant model captures the dynamic behavior of the converter about the operating point and allows derivation of the necessary transfer functions for loop compensation.

3.1 Control-to-Output Transfer Function

The control-to-output transfer function **Tp(s) = V_o(s) / d(s)** describes how the output voltage responds to small variations in the duty cycle, with the input voltage held constant. This transfer function is the fundamental plant model for the outer voltage loop. For the flyback converter, this transfer function contains conjugate pair of LHP poles, one LHP zero , and one RHP zero.

$$Tp(s) = T_{po} * \frac{(1+\frac{s}{\omega_{zn}})(1-\frac{s}{\omega_{zp}})}{(\frac{s}{\omega_o})^2+\frac{2\xi s}{\omega_o}+1}$$

$$\xi = \frac{L_m + C[r(R_L+r_c)+n^2R_Lr_c(1-D)^2]}{2\sqrt{L_mC(R_L+r_c)[n^2R_L(1-D)^2+r]}}$$

$$\omega_{zn} = \frac{1}{r_cC}$$

$$\omega_{zp} = \frac{n^2R_L(1-D)^2 - Dr_L - D^2r_{DS} - n^2D(1-D)R_F}{DL_m}$$

$$\omega_o = \sqrt{\frac{r+n^2R_L(1-D)^2}{L_mC(R_L+r_c)}}$$

Where T_{po} is the DC gain, **ω_zp** is the RHP zero frequency, and **ω_p** is the dominant pole frequency.

The DC and low-frequency gain T_{po} is approximately **19.6523 V (25.869 dB)**. The natural corner frequency formed by the inductor, capacitor, and load is $\boldsymbol{\omega_o}$**= 1.35kHz**, with a damping coefficient $\boldsymbol{\xi}$ **= 0.3698**, causing the gain to roll off at −40 dB/dec beyond this frequency. The converter also exhibits a right-half-plane (RHP) zero at $\boldsymbol{\omega_{zp}}$**= 356.55 * 10^3 rad/s (56.73 kHz)** and a left-half-plane (LHP) zero at $\boldsymbol{\omega_{zn}}$ **= 666.7 * 10^3 rad/s (106.10 kHz).** The RHP zero affects magnitude like a zero but introduces a phase lag similar to a pole, effectively reducing the slope to −20 dB/dec beyond $\boldsymbol{\omega_{zp}}$.

Figure 3.1:*Root Locus of Tp*

3.2 Current-Loop Plant Transfer Function Tpi(s)

The current-loop plant transfer function **Tpi(s) = iL(s) / d(s)** represents the transfer function from duty cycle to inductor current. This is the plant seen by the inner current loop compensator. For the flyback converter, this transfer function is first-order with a single dominant pole.

$$\mathbf{Tpi(s)} = \boldsymbol{T_{pio}} * \frac{\mathbf{1} + \dfrac{\boldsymbol{s}}{\boldsymbol{\omega_{zi}}}}{\left(\dfrac{\boldsymbol{s}}{\boldsymbol{\omega_o}}\right)^2 + \dfrac{\mathbf{2}\boldsymbol{\xi s}}{\boldsymbol{\omega_o}} + \mathbf{1}}$$

$$T_{pio} = \frac{nV_o}{D} \cdot \frac{1 + D}{n^2 R_L (1 - D)^2 + r}$$

$$\omega_{zi} = \frac{1 + D}{C[r_c(1 + D) + R_L]}$$

The DC and low-frequency gain is $\boldsymbol{T_{pio}}$ **= 1.24A (1.87 dB)**. The corner frequency is $\boldsymbol{\omega_o}$**= 1.34Khz**. The transfer function includes a left-half-plane (LHP) zero $\boldsymbol{\omega_{zi}}$ **= 526.250 rad/sec (83.7Hz)** and a pair of complex-conjugate poles at $\boldsymbol{\omega_o}$.

3.3 RHP Zero and Pole-Zero Analysis

The Right-Half-Plane (RHP) zero is a fundamental characteristic of the flyback converter in CCM. It arises because an increase in duty cycle initially stops the conduction of diode, temporarily decreasing the output current and hence the output voltage before the steady-state output rises. This non-minimum-phase behavior introduces a 90° phase lag at the RHP zero frequency, severely constraining the achievable outer voltage loop bandwidth.

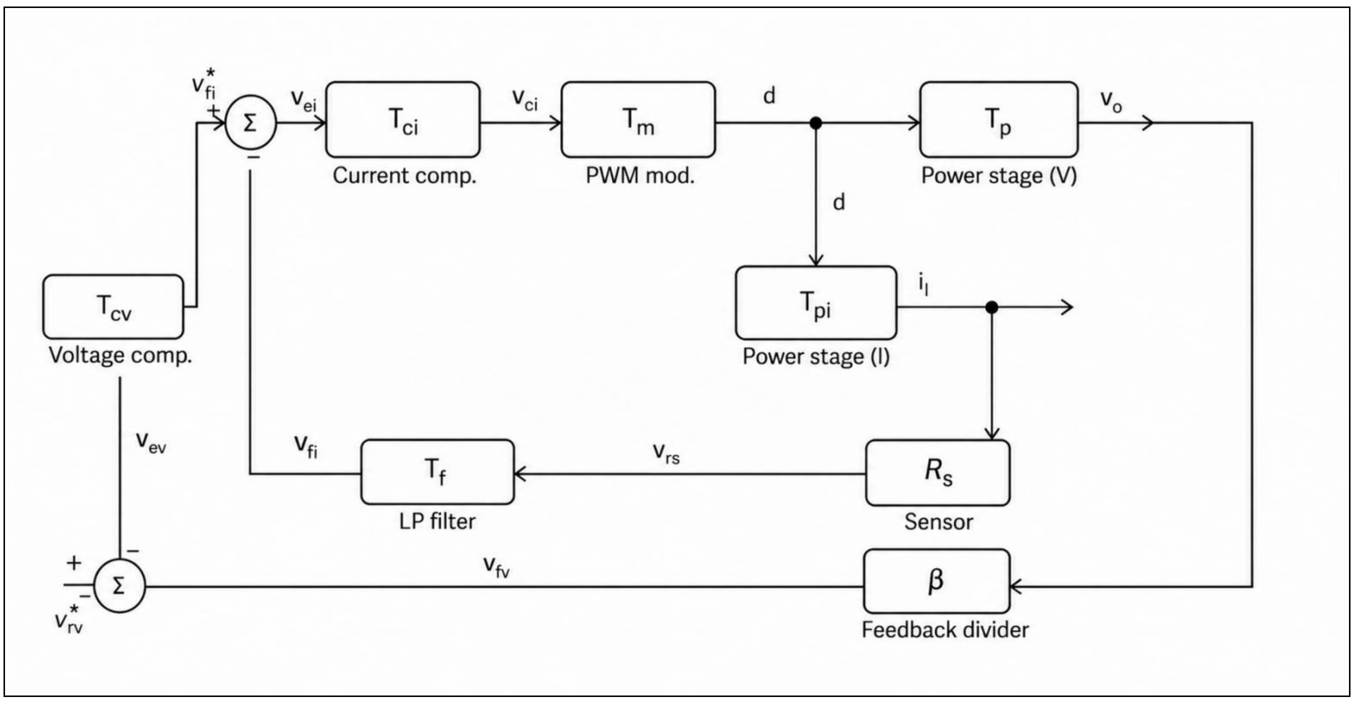
For our design under consideration, the RHP zero frequency is located at: ω_{zp} = 356.55 * 10^3 rad/s (56.73 kHz)

4. Average Current Mode Control Architecture

4.1 Principle of Average Current Mode Control

Average Current Mode Control (ACMC) is a two-loop control strategy in which the inner loop regulates the average value of the inductor current to a reference set by the outer voltage loop. Unlike peak current mode control, ACMC samples and integrates the instantaneous inductor current waveform to obtain its average value, thereby eliminating the subharmonic oscillation problem inherent at duty cycles exceeding 50% in peak CMC. The average current is fed back through a current error amplifier whose output drives the PWM modulator, closing the inner current loop.

The outer voltage loop compares the sampled output voltage with a reference voltage and generates the current reference for the inner loop. Because the inner current loop bandwidth is set significantly higher than the outer voltage loop bandwidth, the inner loop appears as an ideal controlled current source to the outer voltage loop, simplifying the outer loop plant.



4.2 Inner Current Loop Architecture

The inner current loop consists of: (i) a current sensing network with gain Rs , which converts the inductor current into a voltage followed by a low pass filter T_f (ii) a inner loop Compensator with transfer function Tci; and (iii) a PWM modulator with gain ($\mathbf{T_m} = \frac{\mathbf{1}}{\boldsymbol{V_m}}$), where (Vm) is the peak-to-peak amplitude of the sawtooth signal. For simplicity in design and to avoid additional delay, low pas filter is avoided in the simulation of the circuit which is generally placed to reduce the ripples in the inductor current . Moreover, Vm is assumed to be 1, which gives (Tm = 1). The open-loop gain of the inner current loop is therefore:

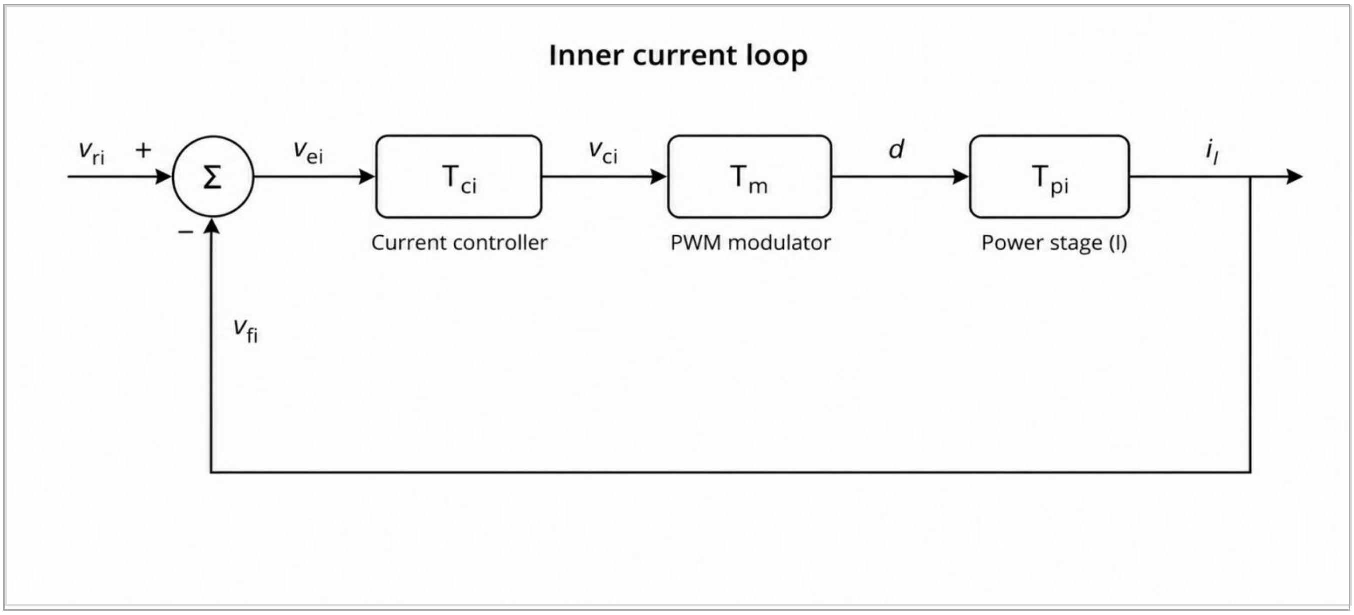
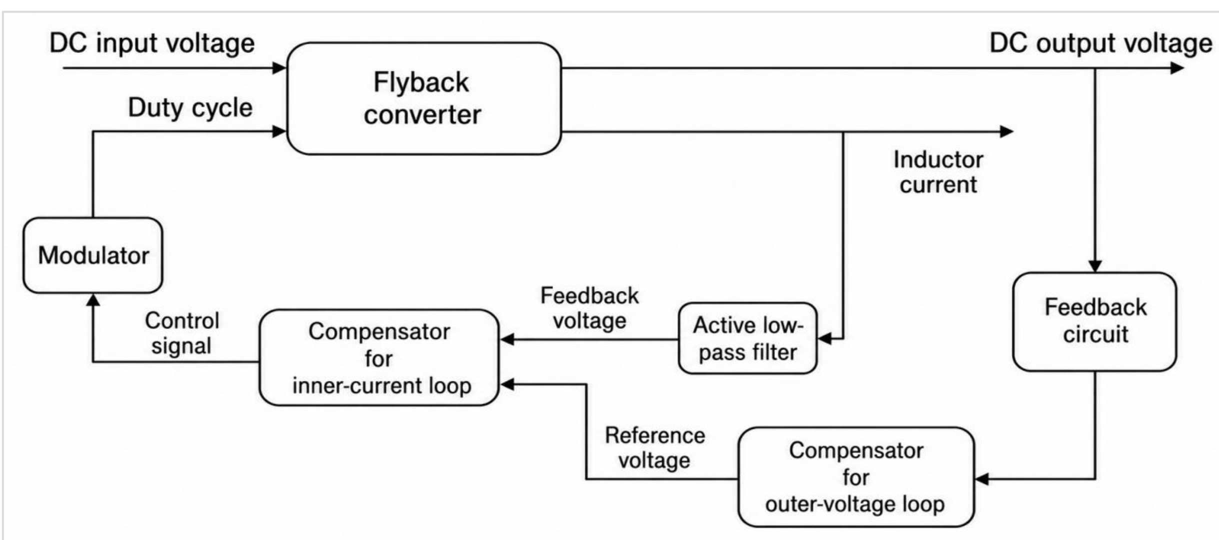


Figure 4.1: *Block Diagram of the inner loop Control Architecture for the CCM Flyback Converter.*

4.3 Outer Voltage Loop Architecture

The outer-voltage loop regulates the flyback converter's output voltage against load disturbances by comparing a fraction of the output voltage (via voltage sensing gain) to reference voltage **Vrv** through the voltage error amplifier. It comprises: (i) the closed inner-current-loop plant (from current reference $\mathbf{V_{RI}}$ to output voltage), (ii) the sensing network, and (iii) the VEA. The loop generates $\mathbf{V_{RI}}$ for the fast inner-current loop using the closed-loop power stage ($\mathbf{V_{RI}}$ to duty cycle d) as its plant, maintaining slower bandwidth to ensure stable output voltage while the inner loop handles rapid inductor current changes.



4.4 Bandwidth Separation and Crossover Selection

The two-loop design relies on adequate bandwidth separation between the inner and outer loops. A ratio of at least 5:1 to 10:1 between the inner current loop crossover frequency **fc** and the outer voltage loop crossover frequency **fcv** is typically maintained. This ensures that the inner loop dynamics do not significantly interact with the outer loop within the outer loop bandwidth.

Given the RHP zero is at = **56.73 kHz**, the outer voltage loop crossover frequency is selected as **3KHz**. Accordingly, the inner current loop crossover frequency is selected as **fc = 20KHz**, which also satisfies fc = fsw/5 = 20 kHz to maintain switching frequency noise rejection.

5. Inner Current Loop Compensator Design

5.1 Design Targets and Specifications

The inner current loop compensator is designed to satisfy the following specifications:

- Inner loop crossover frequency: fc = 20KHz
- Phase margin: PM = (45-60)°
- PWM modulator gain: Tm = 1/Vm = 1.

5.2 Compensator Design Procedure

The current compensator is designed as a Type-II compensator to provide adequate phase boost at the desired crossover frequency while attenuating switching ripple at higher frequencies.

i. **The Uncompensated Loop Gain T_{ki}(s)**

$$T_{ki} = T_m T_{pi}$$

$$T_m = 1$$

$$T_{ki} = T_{pi}$$

$$\boxed{T_{ki} = \frac{7.727\times10^{11}s+4.066\times10^{14}}{4.48\times10^{6}s^{2}+2.821\times10^{10}s+3.247\times10^{14}}}$$

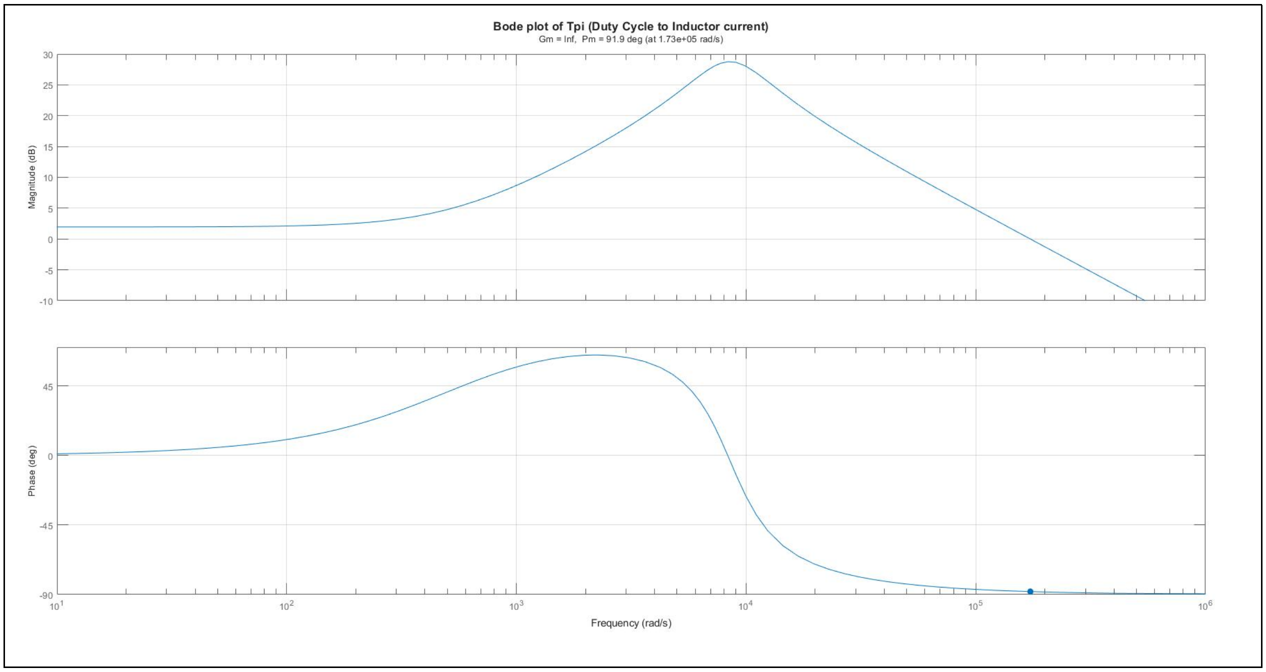


Figure 5.1:*Bode of Uncompensated inner loop {Tki}*

ii. Transfer Function of Compensation Circuit T_{ci} (s)

The desired crossover frequency for the loop gain is chosen as **fc = 20 kHz**. At this frequency, the magnitude of the **uncompensated loop gain transfer function** is $|T_{ki}(f_c)| = 1.3770$, and its phase is $\phi_{T_{ki}}(f_c) = -87.35°$**.** The required phase margin is **PM = 60 ,** Therefore, the phase boost required at **fc** is

$$\phi_m = PM - \phi_{T_k}(f_c) - 90° = 57.35°$$

$$K = \tan\left(\frac{\phi_m}{2} + 45°\right) = 3.414$$

$$\omega_c = \sqrt{\omega_{zci}\,\omega_{zpi}}\ \ , \qquad K = \sqrt{\frac{\omega_{pci}}{\omega_{zi}}}$$

On solving we get the values of $\omega_{zci} = 36.796 * 10^3$, $\omega_{zpi} = 429.17 * 10^3$ **rad/sec**

$$T_{ci} = T_{ci0}\,\frac{1 + \frac{s}{\omega_{zci}}}{s\left(1 + \frac{s}{\omega_{pci}}\right)}$$

$$T_{ci0} = 2.6721\text{e} + 04$$

$$\text{Tci} = \frac{1.147 \times 10^{10}s + 4.22 \times 10^{14}}{3.68 \times 10^{4}s^{2} + 1.579 \times 10^{10}s}$$

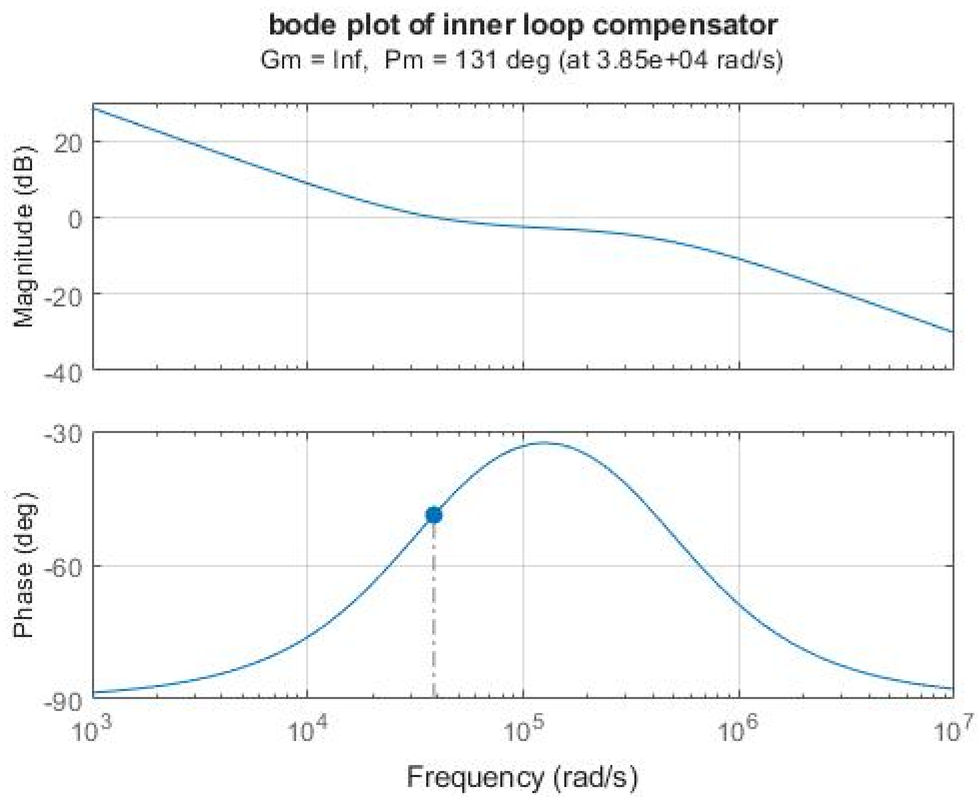


Figure 5.2:*Bode plot of inner Type2 compensator Tci(s)*

iii. Compensated Loop Gain T_i

$$T_i = T_{ki}\,T_{ci} = T_m\,T_{pi}\,T_{ci}$$

$$\text{Ti} = \frac{8.861 \times 10^{21}s^{2} + 3.307 \times 10^{26}s + 1.716 \times 10^{29}}{1.648 \times 10^{11}s^{4} + 7.179 \times 10^{16}s^{3} + 4.573 \times 10^{20}s^{2} + 5.127 \times 10^{24}s}$$

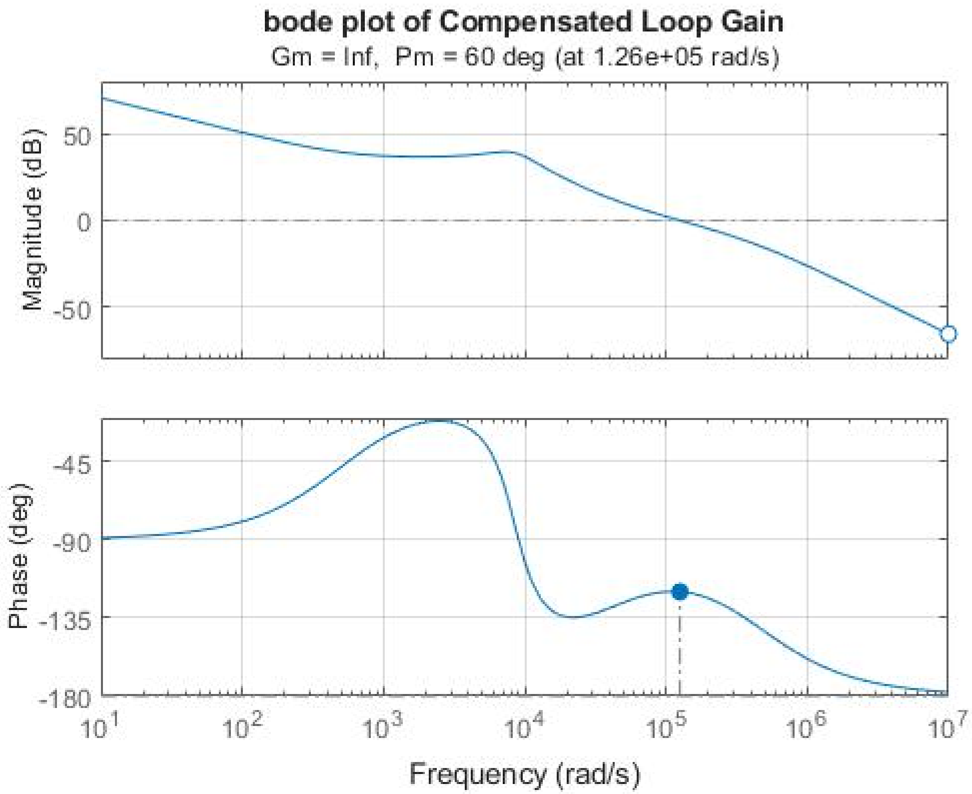


Figure 5-3:Bode of Compensated LOOP Gain

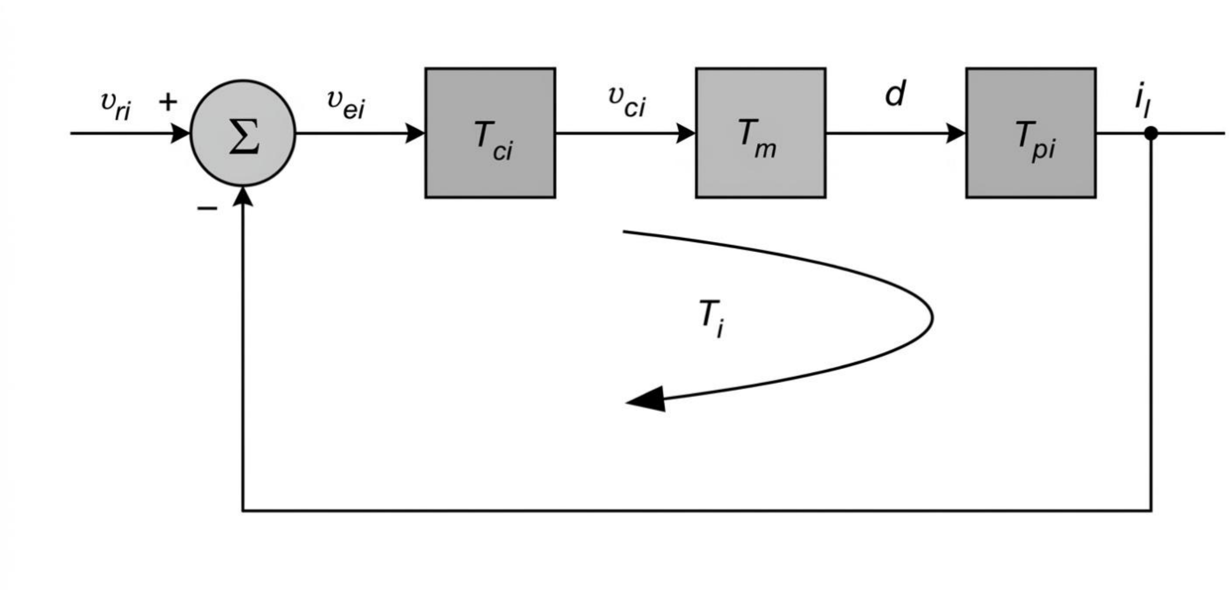
5.4 Gain Margin and Phase Margin Verification

The gain and phase margins of the compensated inner current loop are tabulated below. Both margins satisfy the design targets, confirming adequate stability margin.

Parameter	Target	Achieved
Crossover Frequency f_ci	20K Hz ($1.25*10^5$ rad/s)	20.053KHz ($1.26*10^5$ rad/s)
Phase Margin PM_i	≥ 45°	60°

6. Closed-Loop Transfer Functions for Inner-Current Loop

6.1 Reference voltage to inductor current transfer function(Ticl)



$$\boldsymbol{T_{icl}(s) = \frac{i_l(s)}{v_{ri}(s)}\bigg|_{v_i=i_o=0} = \frac{T_{ci}T_mT_{pi}}{1+T_{ci}T_mT_{pi}}}$$

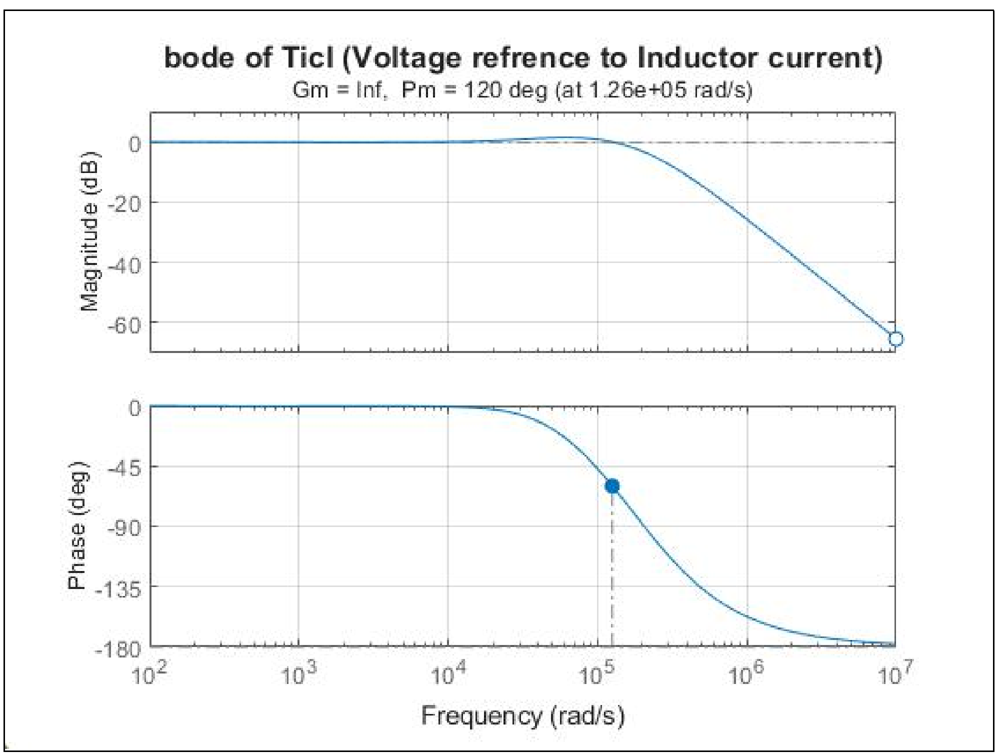


Figure 6-1:Bode of Inductor current to Refrence voltage ***Ticl***

6.2 Reference Voltage-to-Output Voltage Transfer Function Tpicl(s)

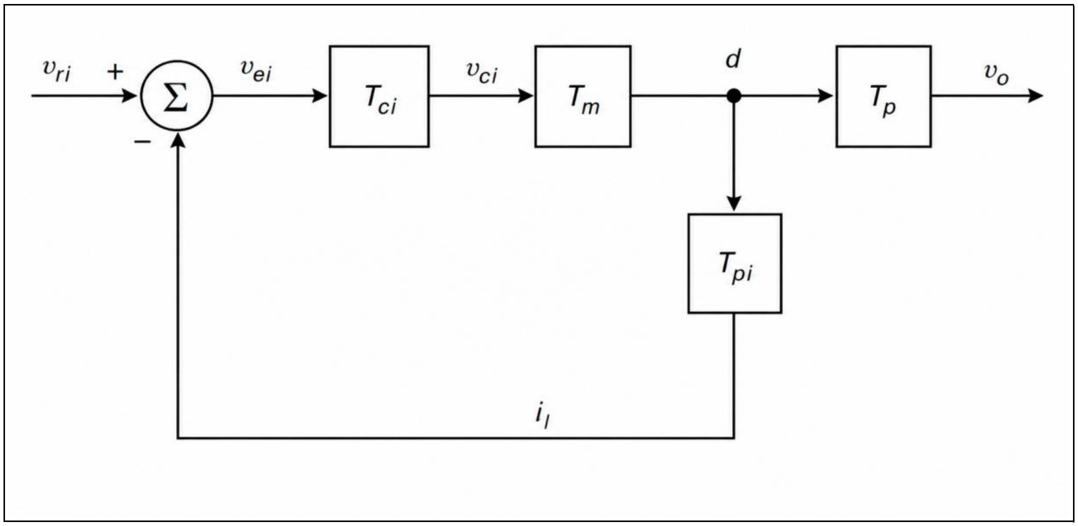
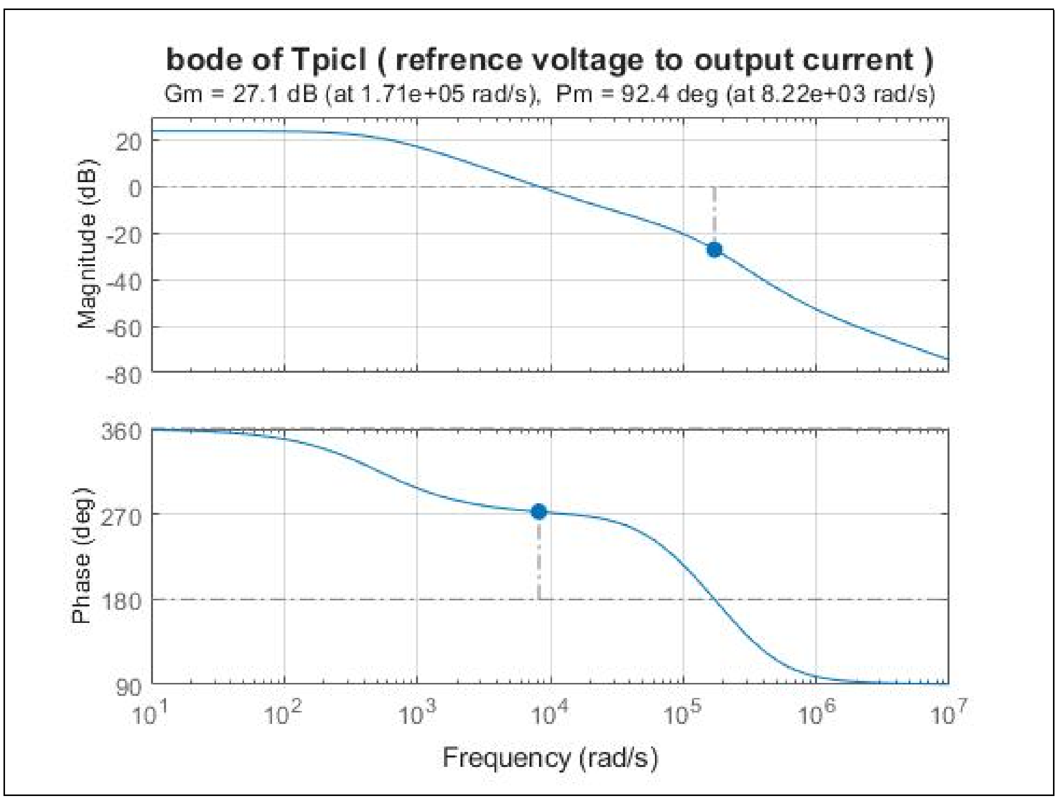


*Figure 6-2:Block diagram required to derive the transfer function **Tpicl**.*

$$\boldsymbol{T_{picl}(s) = \left.\frac{v_o(s)}{v_{ri}(s)}\right|_{v_i=i_o=0} = \frac{T_{ci}T_mT_p}{1+T_i}}$$



7. Outer-Voltage Loop

7.1 Uncompensated Loop gain Tkv

$$\boldsymbol{T_{kv} = \beta T_{picl} \quad , \quad \beta=1}$$

The gain is low at DC and is nearly Tkvo = 0.986. The crossover frequency is nearly 1.32KHz,It is essential to provide a high gain at DC to reduce the steady-state error. The crossover frequency must be high for improved bandwidth; however, it cannot be higher than the frequency of the RHP zero present in the Tp transfer function.

7.2Transfer Function of Compensation Circuit Tcv

A detailed procedure to determine the transfer function of the controller to improve the DC gain and achieve an inner-current-loop phase margin PM = 60,From the uncompensated loop gain transfer function Tkv, the crossover frequency is fc = 1.32 KHz. At this frequency, the magnitude of $\boldsymbol{T_{kv}}$(fc) = 0.998 = −0.017 dB, and the phase is −87.6076 degree.

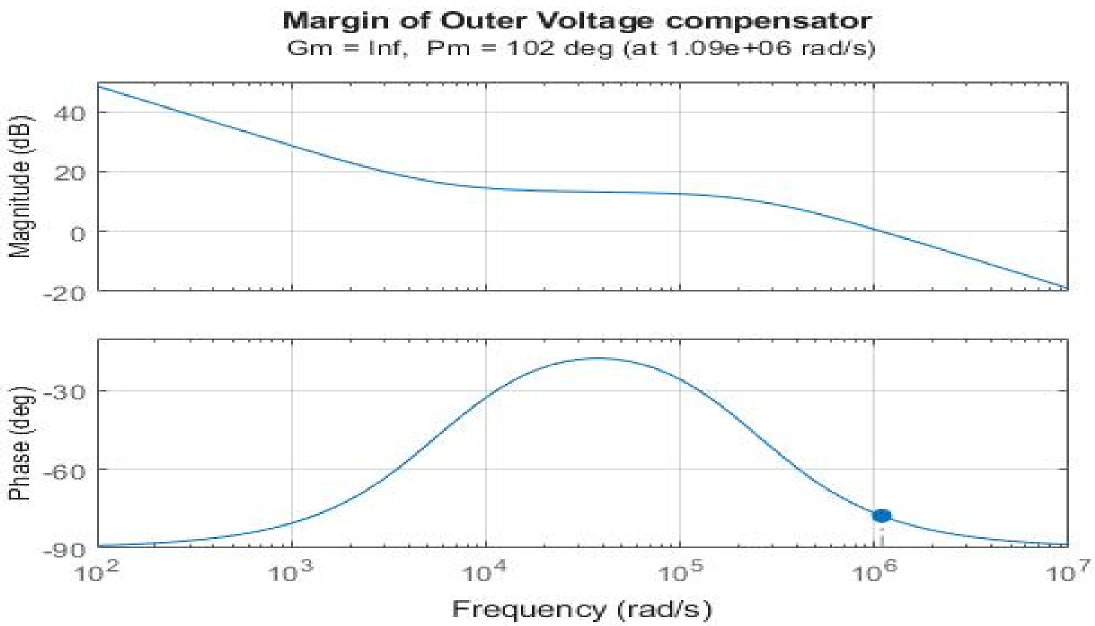
◦ .

$$\boldsymbol{\phi_m = PM - \phi_{T_{kv}}(f_c) - 90^\circ = 57.60}$$

$$\boldsymbol{K = \tan\left(\frac{\phi_m}{2} + 45^\circ\right) = 3.442}$$

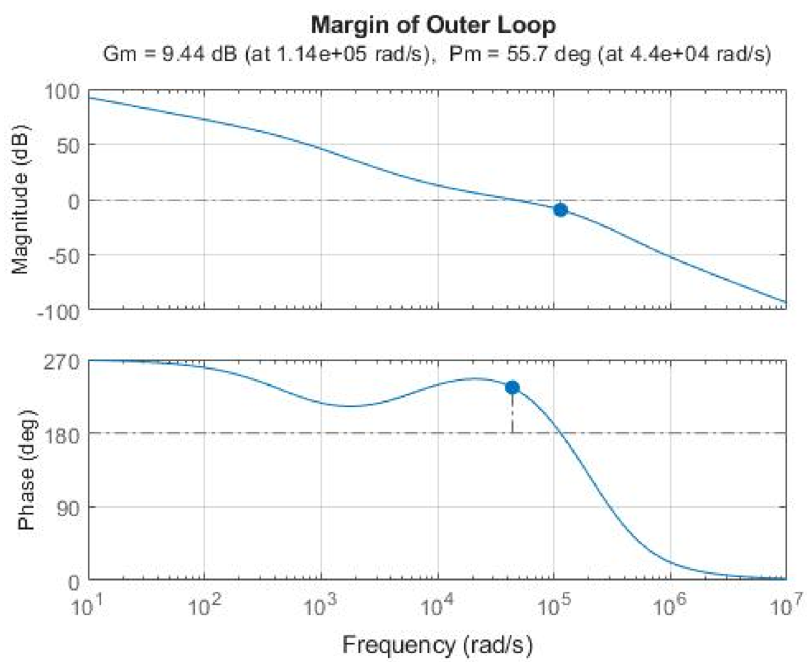
$$\omega_c = \sqrt{\omega_{zci}\,\omega_{zpi}} \quad , \qquad K = \sqrt{\frac{\omega_{pci}}{\omega_{zi}}}$$

$$\boldsymbol{T_{cv} = \frac{7.63 \times 10^8 s + 1.838 \times 10^{12}}{2409 s^2 + 6.879 \times 10^7 s}}$$



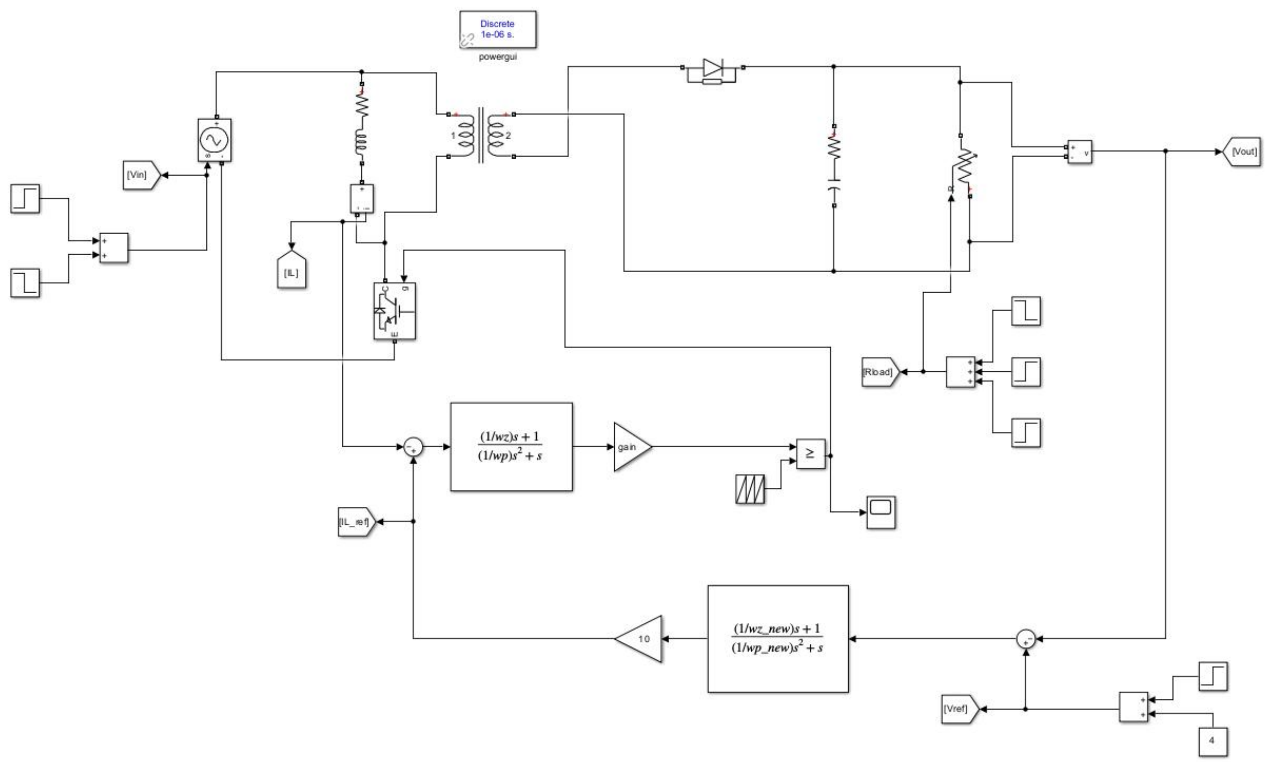
7.2 Compensated Loop Gain Tv

$$T_v = T_{kv}T_{cv} = T_{picl}T_{cv}$$



7. Simulation Results and Discussion

The MATLAB/Simulink environment is used to validate the theoretical analysis and compensator designs. The simulation model implements the complete switched-mode converter with the designed inner current loop and outer voltage loop compensators.



7.1 Reference Tracking Performance

The closed-loop reference tracking performance is evaluated by applying a step change to the output voltage reference from 4 V to 20V at t = 0.5 s. and from 20V to 6V at 0.8s .The output voltage response, settling time, overshoot, and steady-state error are measured. The system is expected to track the new reference within approx 0.05s with an overshoot of less than 1%.

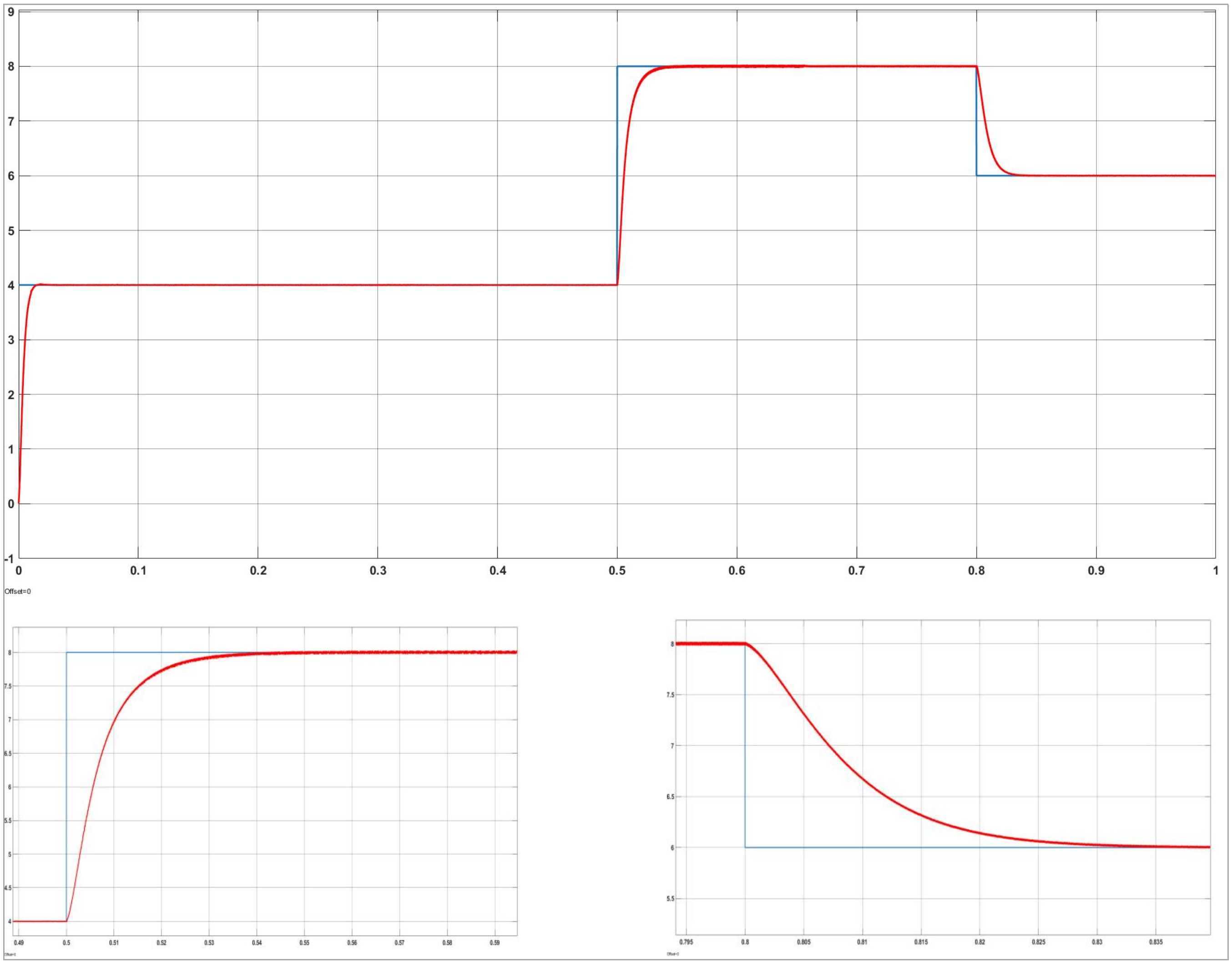


Figure 7.1: *Closed-Loop Output Voltage Reference Tracking Response along with there settling times bottom graphs*

7.4 Line Regulation Response

A step change in the input voltage from 12 V to 16 V is applied at t = 0.3s to evaluate the line regulation performance. The ACMC inner current loop provides inherent input voltage feedforward, significantly attenuating the output voltage deviation caused by input voltage perturbations.

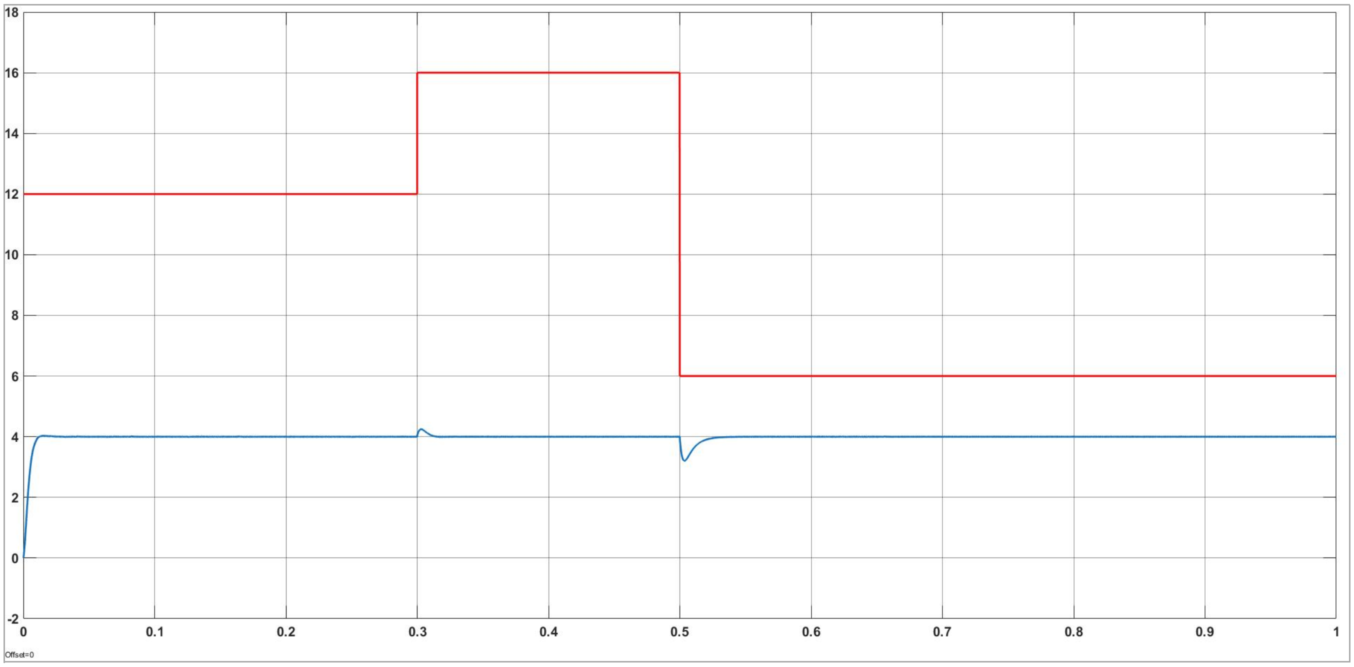


Figure 7.2: *Line Regulation: Output Voltage Response to Input Voltage Step.*

7.5 Load Transient Response

A step change in the load resistance from 20 Ω to 18 Ω at 0.2 sec ,later increasing it by+7 at 0.4sec , is applied to evaluate the load transient response.

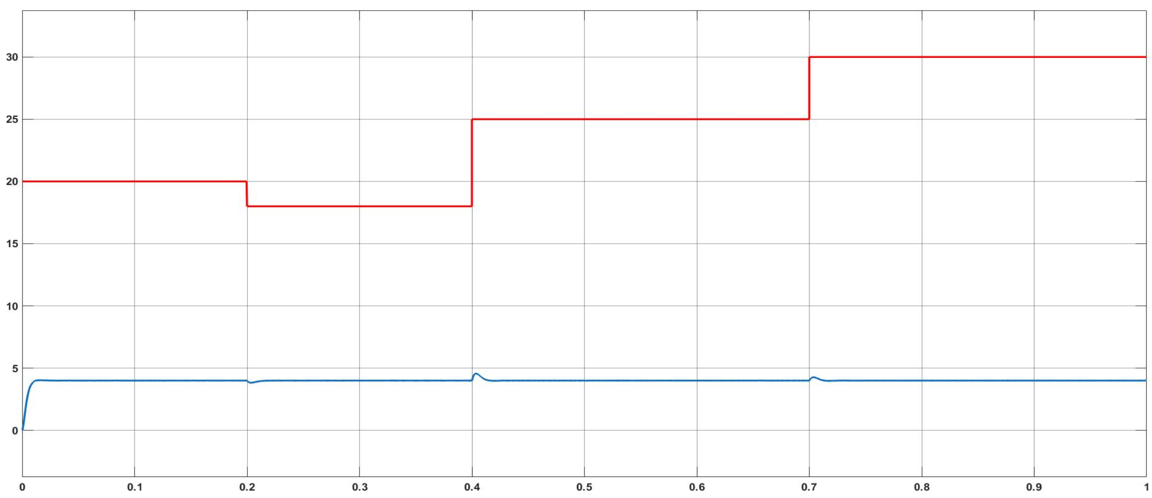


Figure 7.3: *Output Voltage Response to Load changes.*

7.6 Inductor Current and Output Voltage Waveforms

The steady-state inductor current and output voltage waveforms under closed-loop ACMC operation are presented. The waveforms confirm CCM operation, with the average inductor current regulated to the reference level and the output voltage ripple within acceptable bounds.

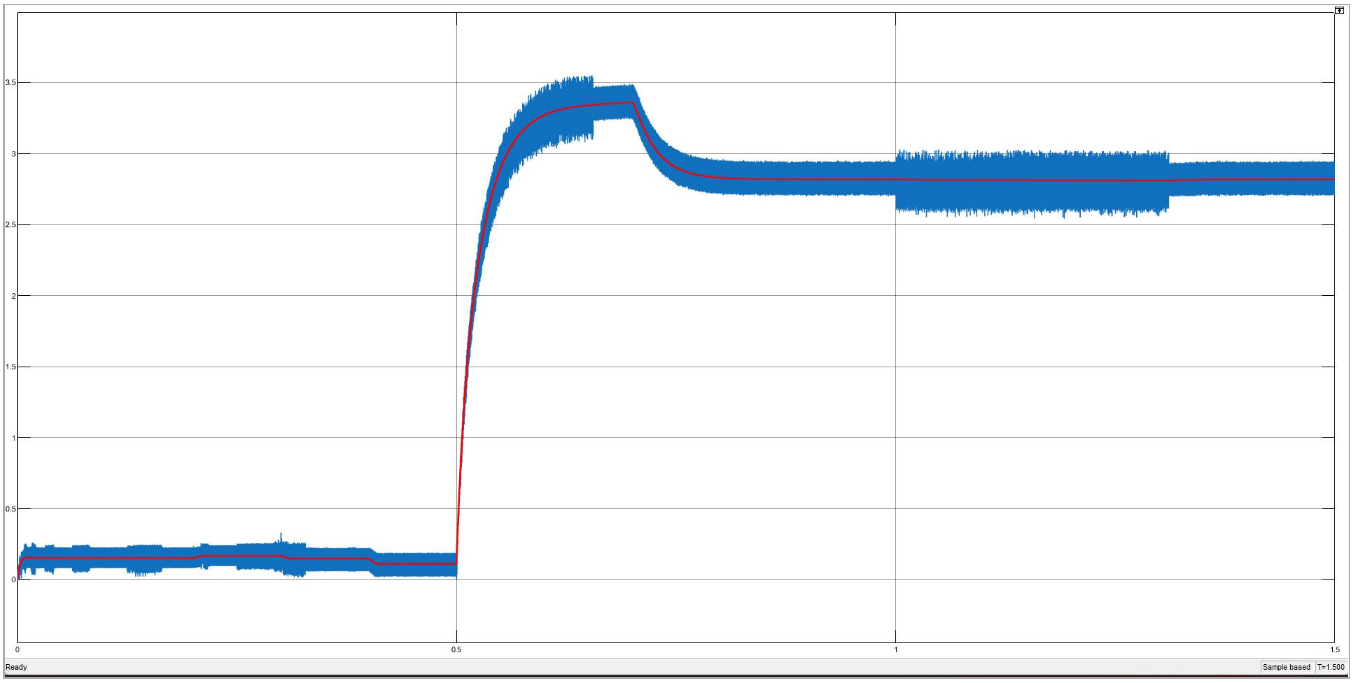


Figure 7.4: *Inductor Current and Control signal*

Signal Statistics

	Value	Time
Max	2.945e+00	1.438
Min	2.703e+00	1.446
Peak to Peak	2.415e-01	
Mean	2.818e+00	
Median	2.818e+00	
RMS	2.818e+00	

Figure 7-5*:current statistics for above graph*

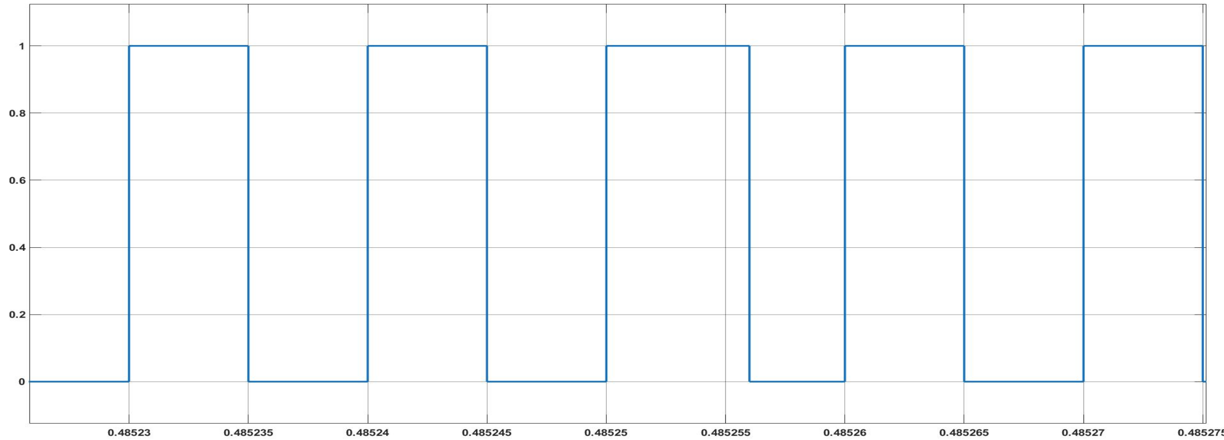


Figure 7-6:*Gate pulse*

The RHP zero limitation is clearly observable in the load transient response, where the initial output voltage deviation is in the 'wrong' direction before recovering to the regulated value. This non-minimum-phase behavior is inherent to the flyback converter in CCM and cannot be eliminated by the control design; it can only be managed by constraining the outer loop bandwidth below the RHP zero frequency.

8. Conclusion

8.1 Summary of Work

This project has presented a comprehensive analysis and closed-loop control design for an Average Current Mode Controlled CCM flyback DC–DC converter. Starting from the circuit description and DC model, the control-to-output and current-loop plant transfer functions were obtained using the circuit averaging technique based on energy conservation, and their frequency-domain characteristics were analyzed, with particular attention to the Right-Half-Plane zero that constrains outer loop bandwidth.

A two-loop compensator architecture was designed: the inner current loop compensator ensures tight regulation of the average inductor current at a crossover frequency of 20 kHz with a phase margin of 60°, while the outer voltage loop compensator maintains output voltage regulation at a crossover frequency of 1.32 kHz with a phase margin of 55°, safely below the RHP zero frequency. The design was validated through MATLAB/Simulink simulation, demonstrating satisfactory reference tracking, line regulation, and load transient performance.

8.2 Key Observations

- The ACMC strategy inherently provides input voltage feedforward through the inner current loop, resulting in superior line regulation performance compared to voltage-mode control.
- The RHP zero at 56.73 kHz is the primary bandwidth-limiting factor for the outer voltage loop, restricting the achievable dynamic performance of the converter.
- CCM operation is maintained across the operating range, confirming that the designed inductance Lm = 200 µH is adequate for the given load and switching frequency conditions.

8.3References

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